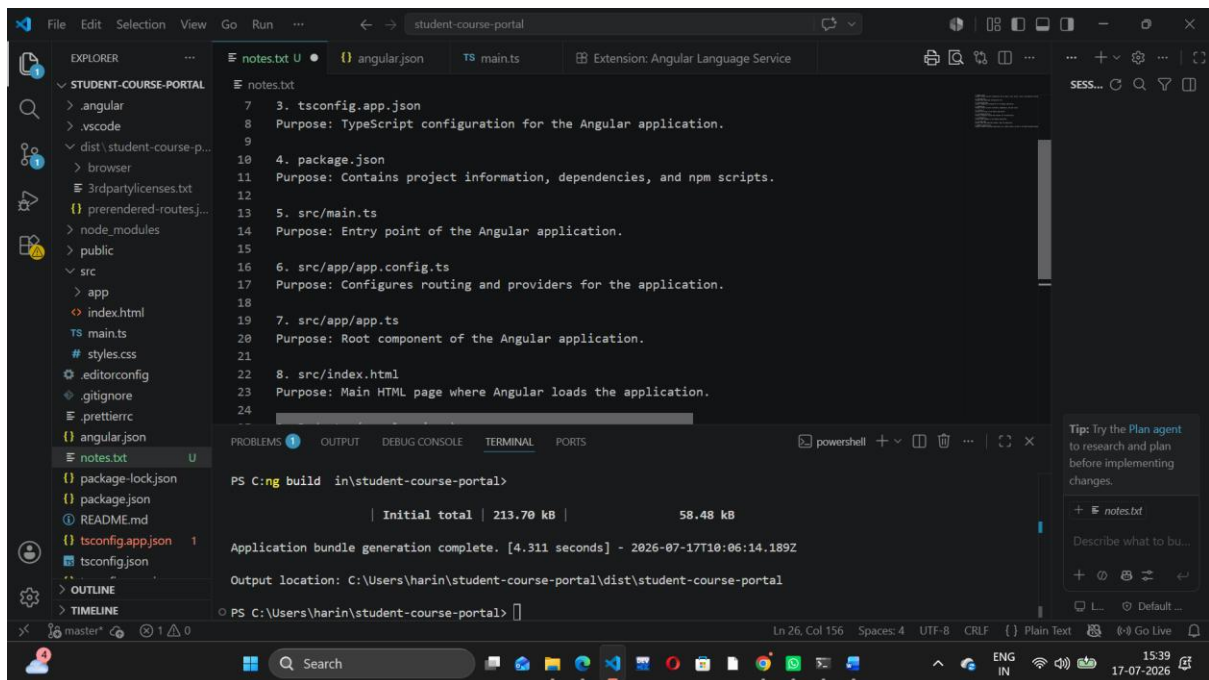


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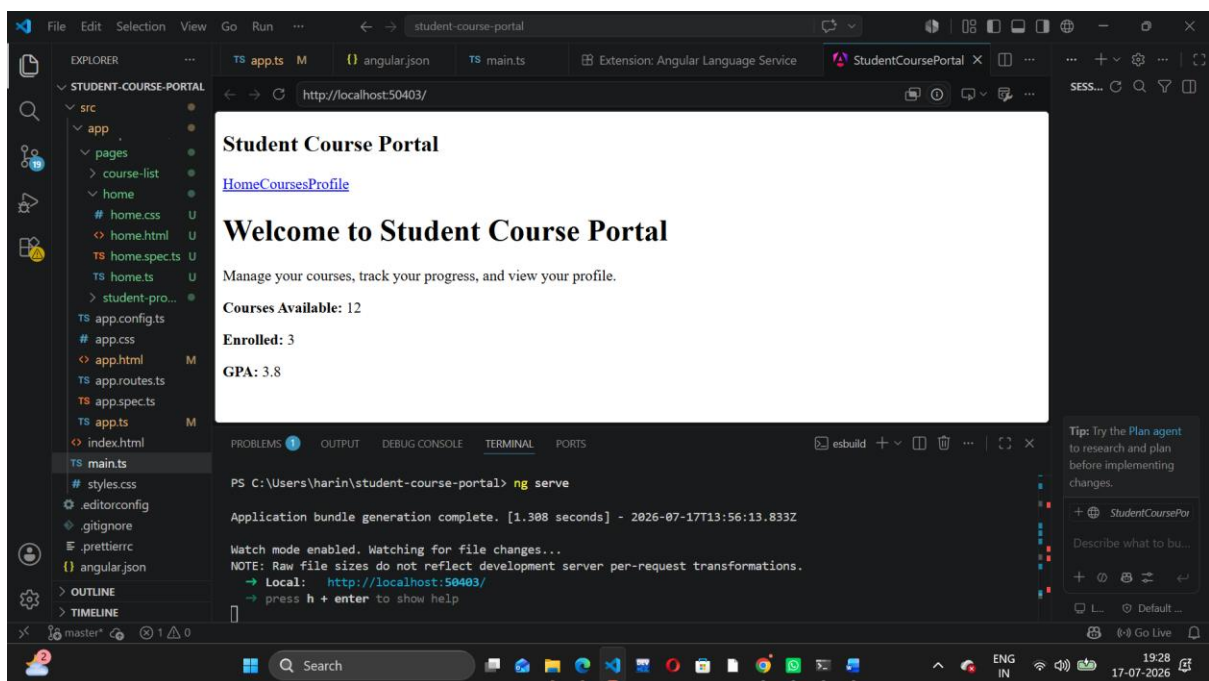
2300100063iot@gmail.com

ANGULAR

Task 1: Scaffold and Explore the Angular Project Goal: Create the Angular workspace and understand what each generated file does

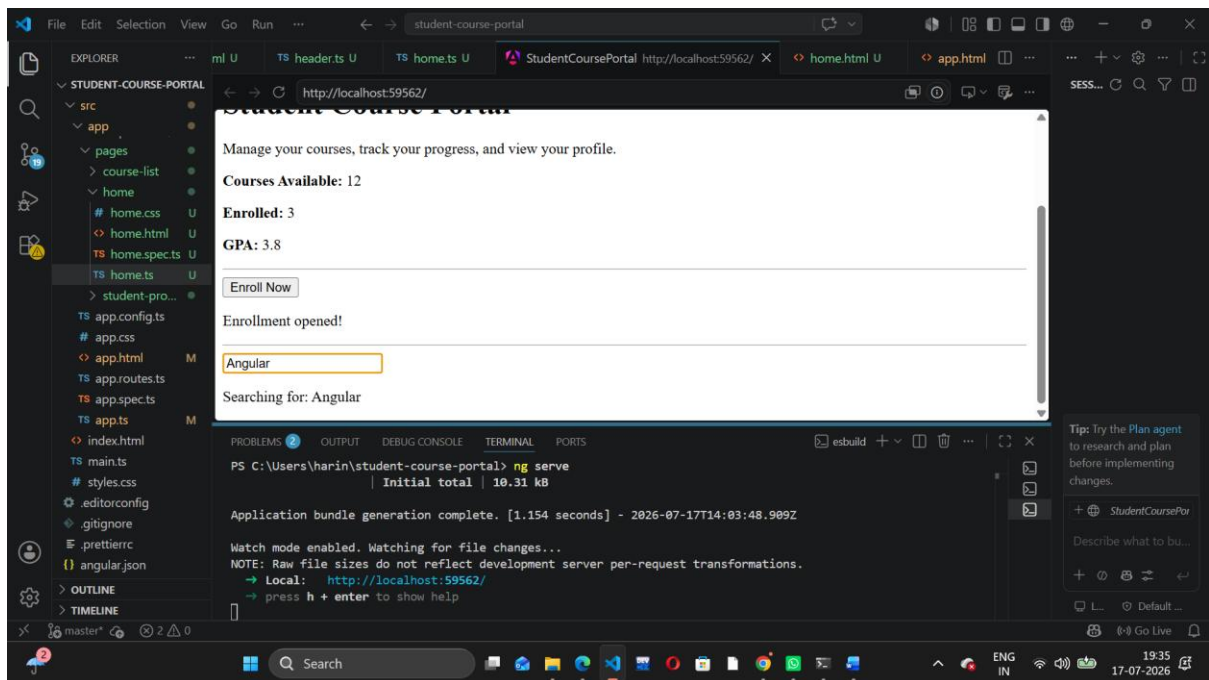


Task 2: Create and Organise Components Goal: Generate the core page components for the Student Course Portal.

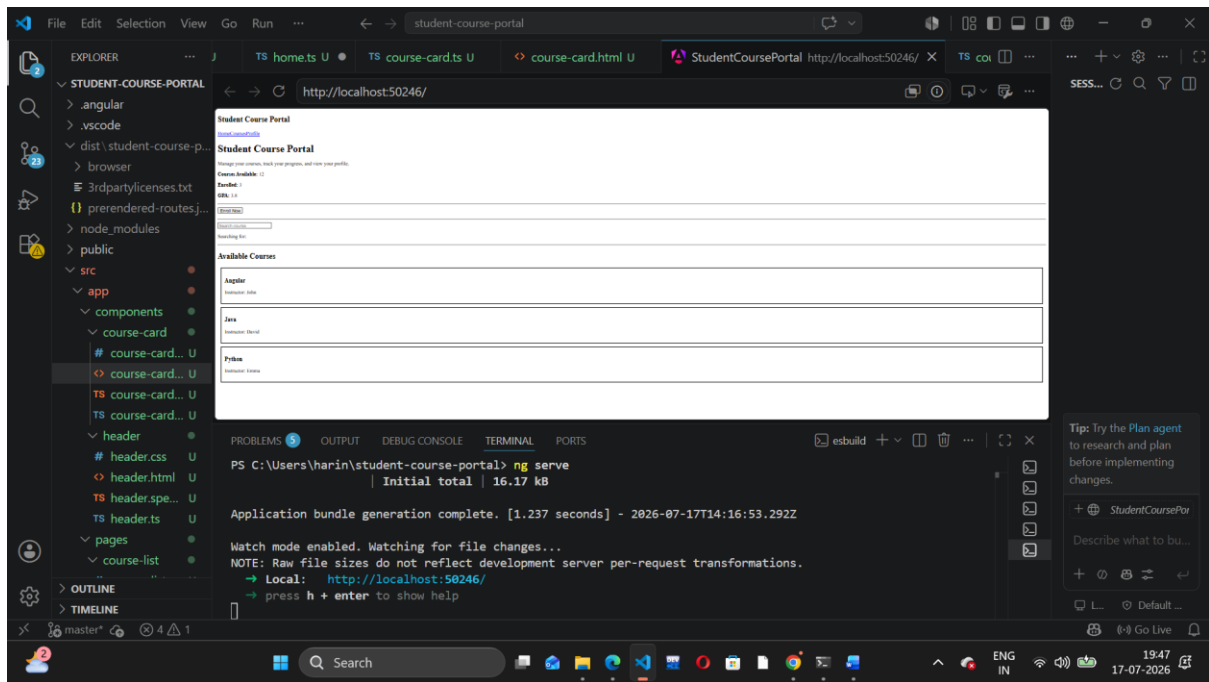


HANDS-ON 2 [Beginner] Digital Nurture 5.0 | Angular (v20.0) | Hands-On Exercise Book ✓ Data Binding, Lifecycle Hooks & Component Communication

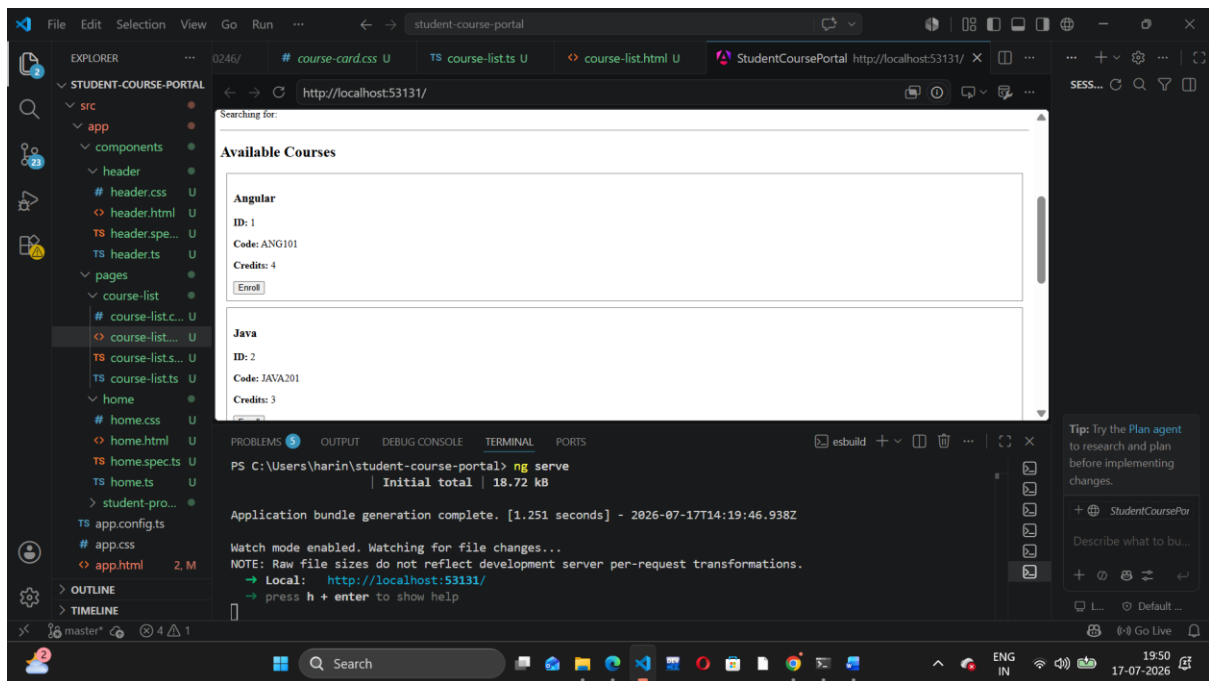
Task 1: All Four Binding Types Goal: Practise interpolation, property binding, event binding, and two-way binding in one component



Task 2: Lifecycle Hooks Goal: Implement the three most important lifecycle hooks and observe when they fire

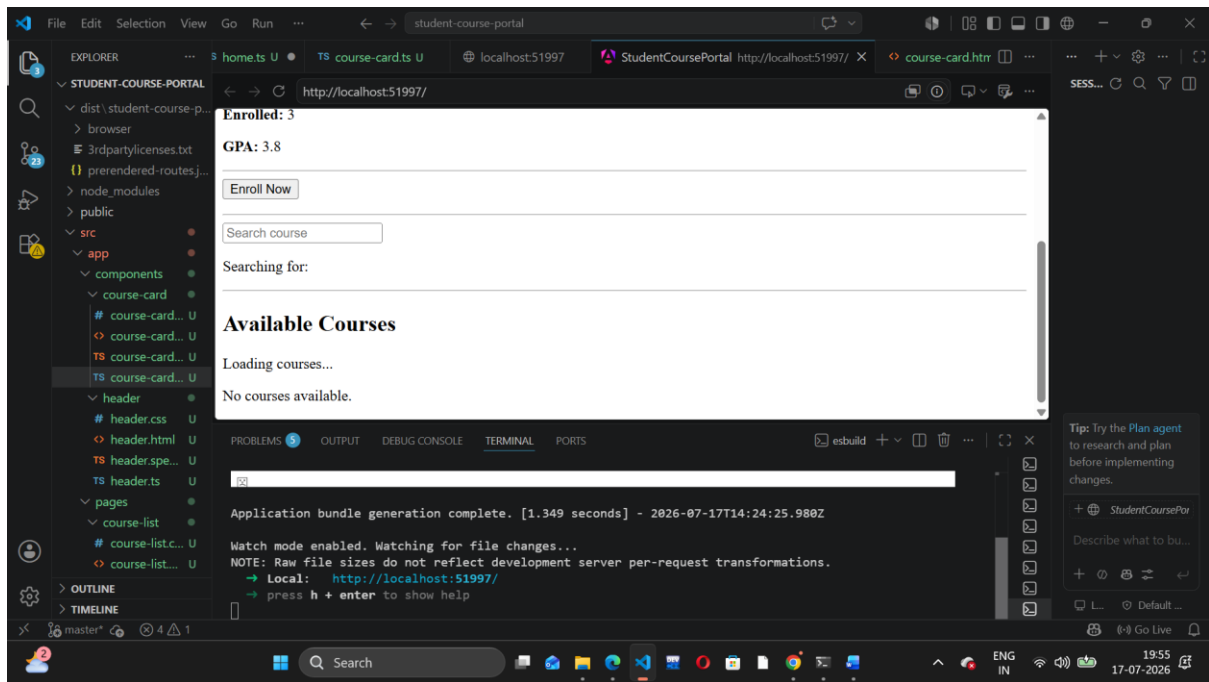


Task 3: @Input and @Output — Parent-Child Communication Goal: Pass data down with @Input and emit events up with @Output

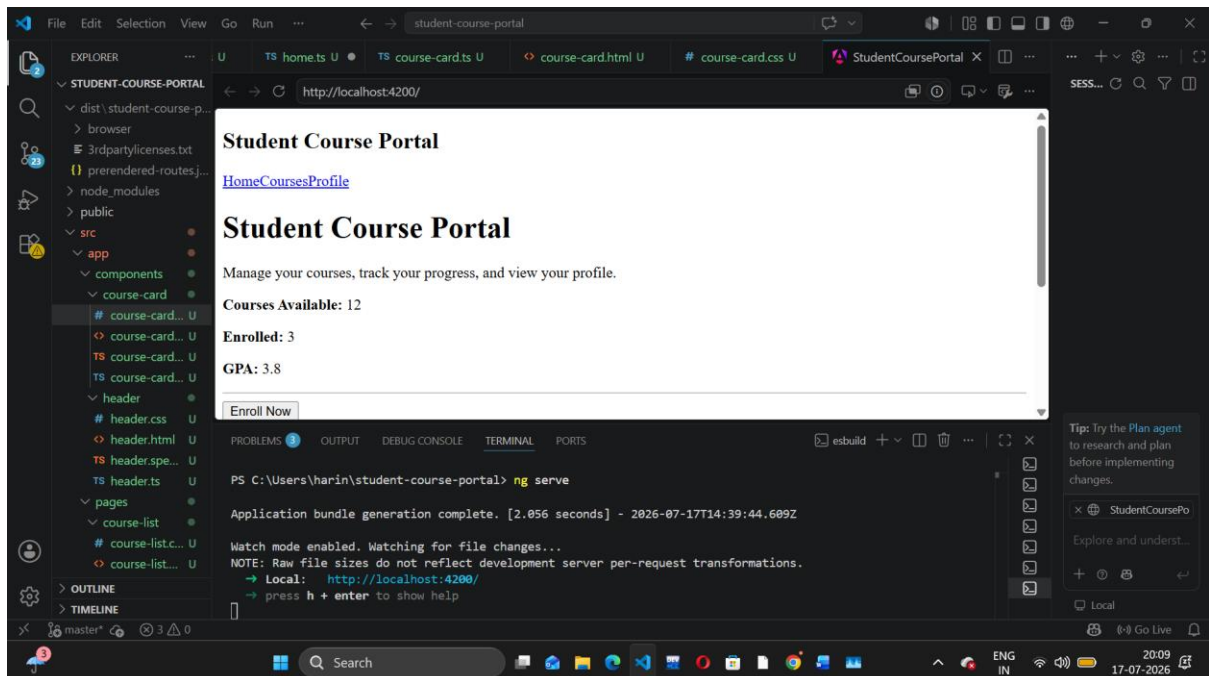


HANDS-ON 3 [Beginner] Digital Nurture 5.0 | Angular (v20.0) | Hands-On Exercise Book ✓ Directives & Pipes — Built-in and Custom

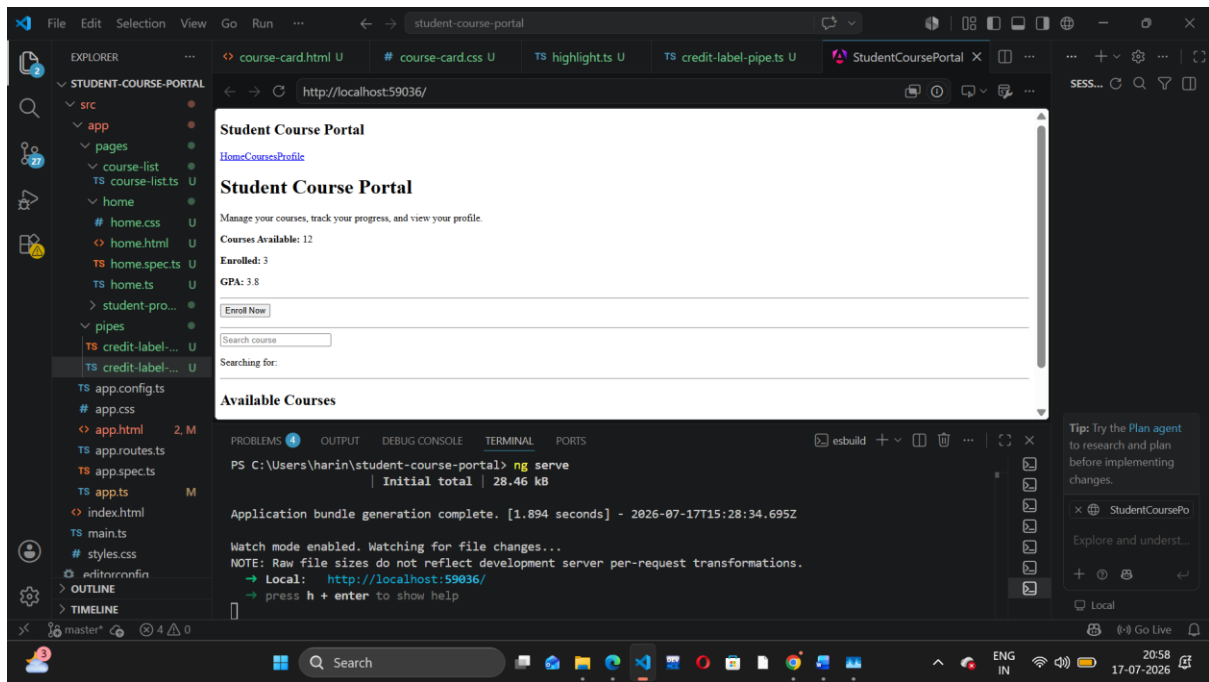
Task 1: Structural Directives Goal: Control DOM structure dynamically using *ngIf, *ngFor, and *ngSwitch.



Task 2: Attribute Directives — ngClass and ngStyle Goal: Apply dynamic CSS classes and inline styles using attribute directives.

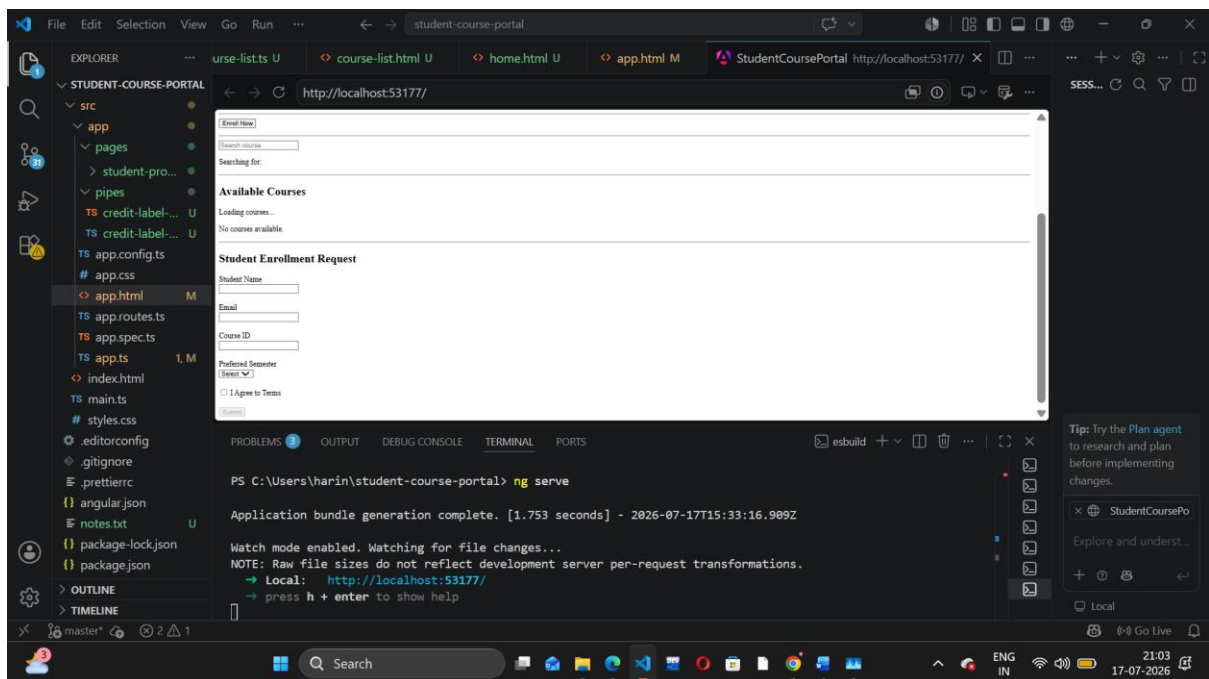


Task 3: Custom Directive and Custom Pipe Goal: Build a reusable attribute directive and a custom pipe for the portal.

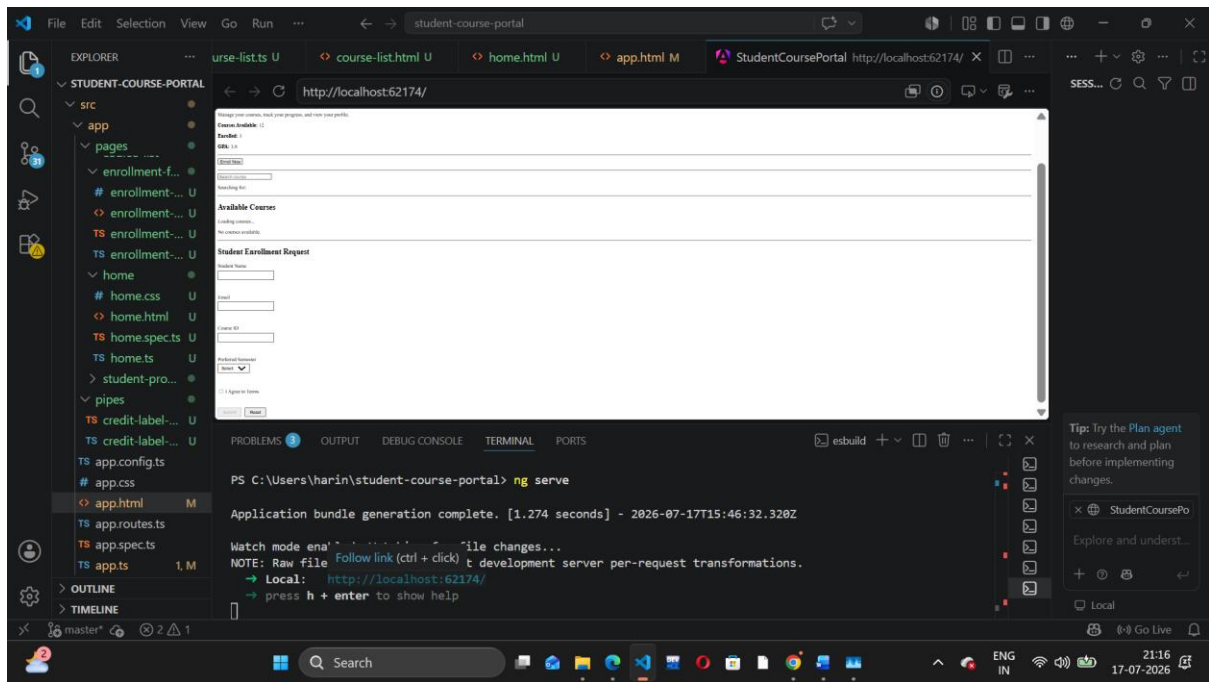


HANDS-ON 4 [Intermediate] Digital Nurture 5.0 | Angular (v20.0) | Hands-On Exercise Book ✓ Template-Driven Forms & Validation

Task 1: Build the Enrollment Request Form Goal: Create a template-driven form with multiple input types and ngModel bindings

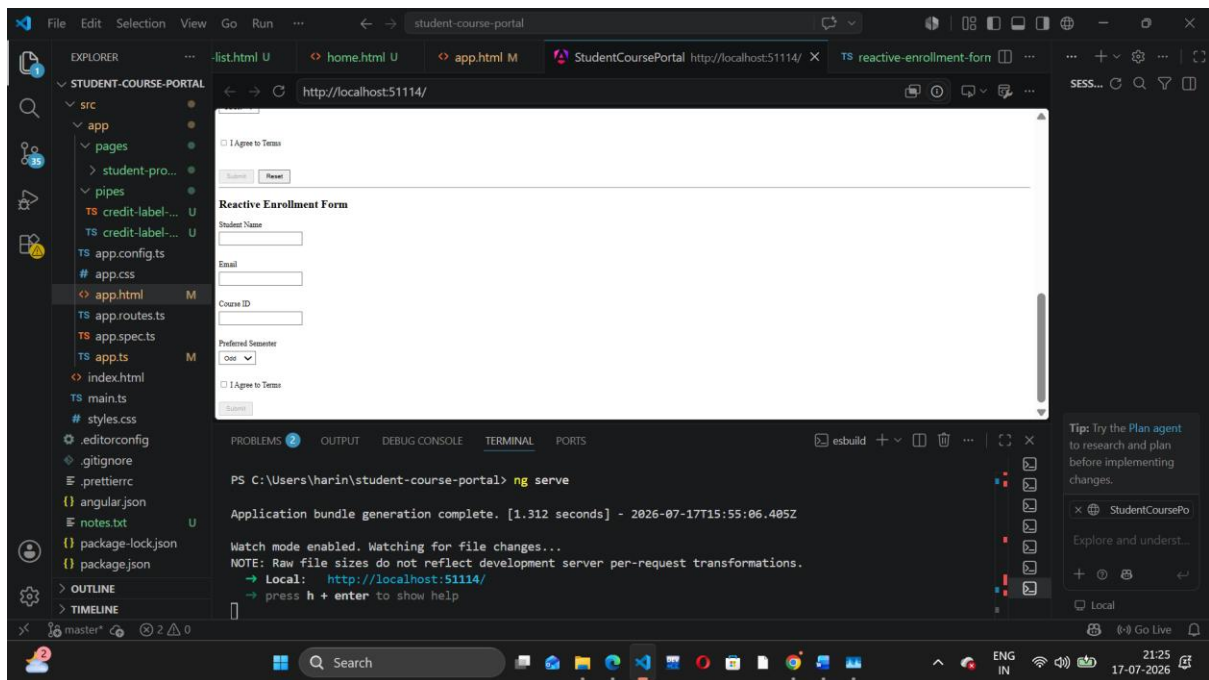


Task 2: Validation and Error Messages Goal: Add built-in validators and display contextual error messages for each field

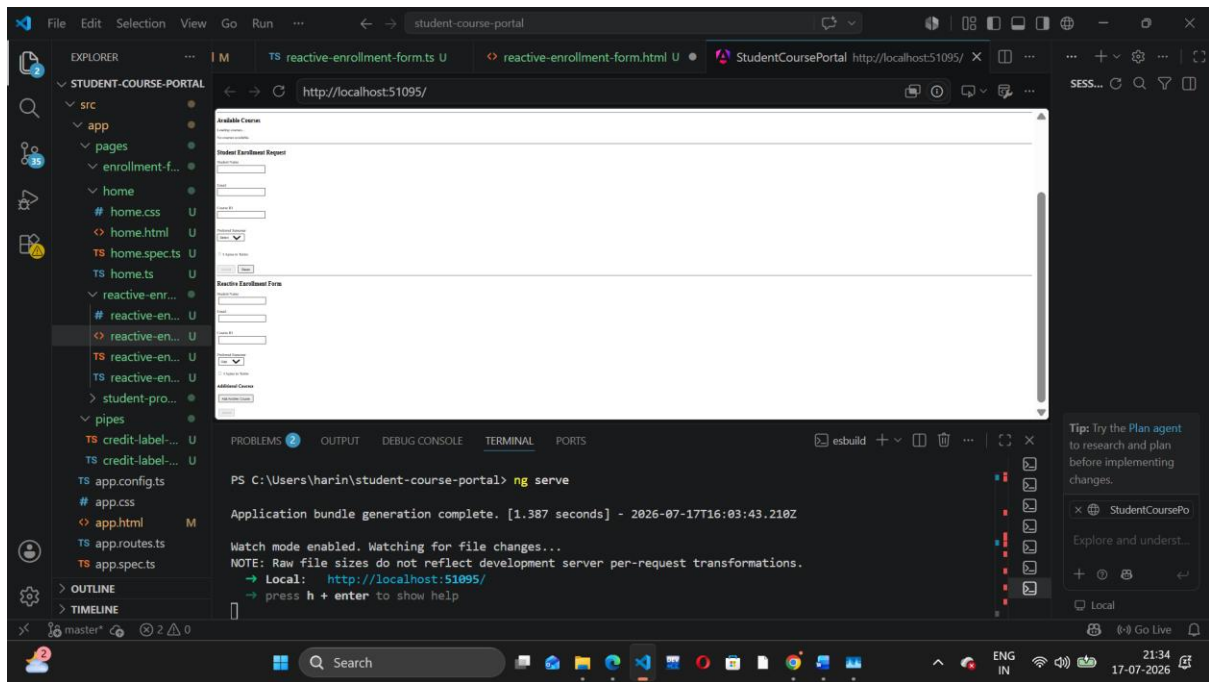


HANDS-ON 5 [Intermediate] Digital Nurture 5.0 | Angular (v20.0) | Hands-On Exercise Book Reactive Forms — FormBuilder, FormGroup, FormArray & Custom Validators

Task 1: Build a Reactive Form with FormBuilder Goal: Define and bind a reactive form using FormBuilder

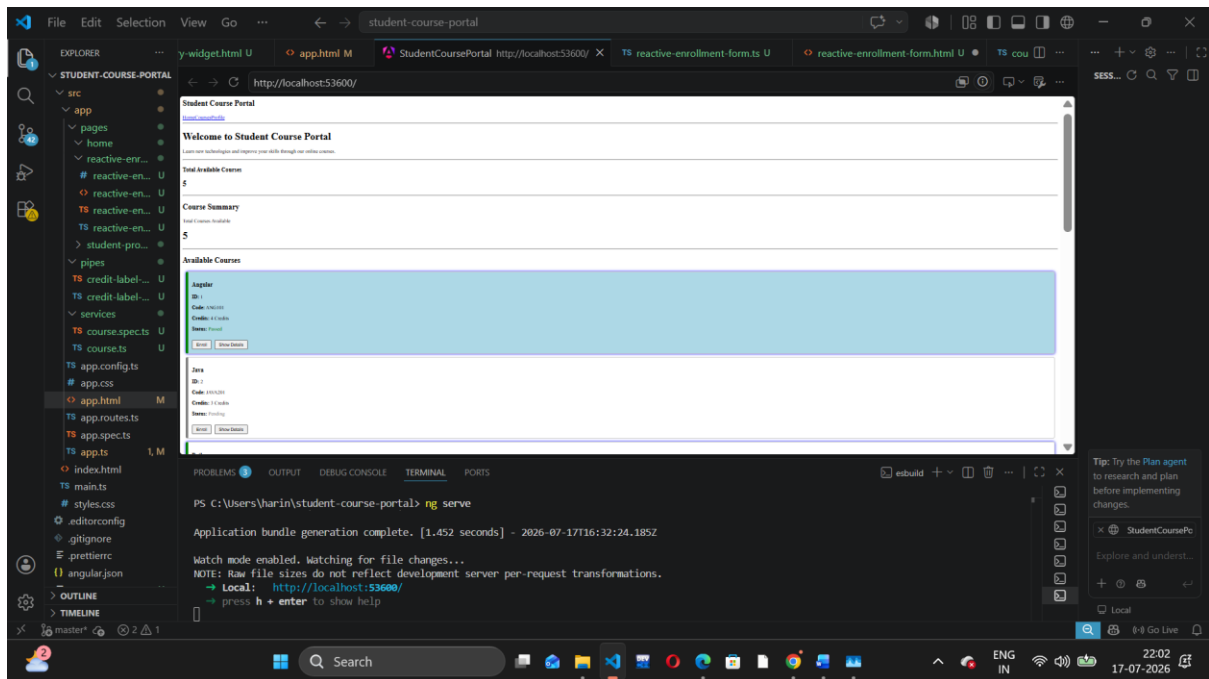


Task 2: Custom Validators and FormArray for Dynamic Controls Goal: Write a custom synchronous validator and use FormArray for repeating controls.



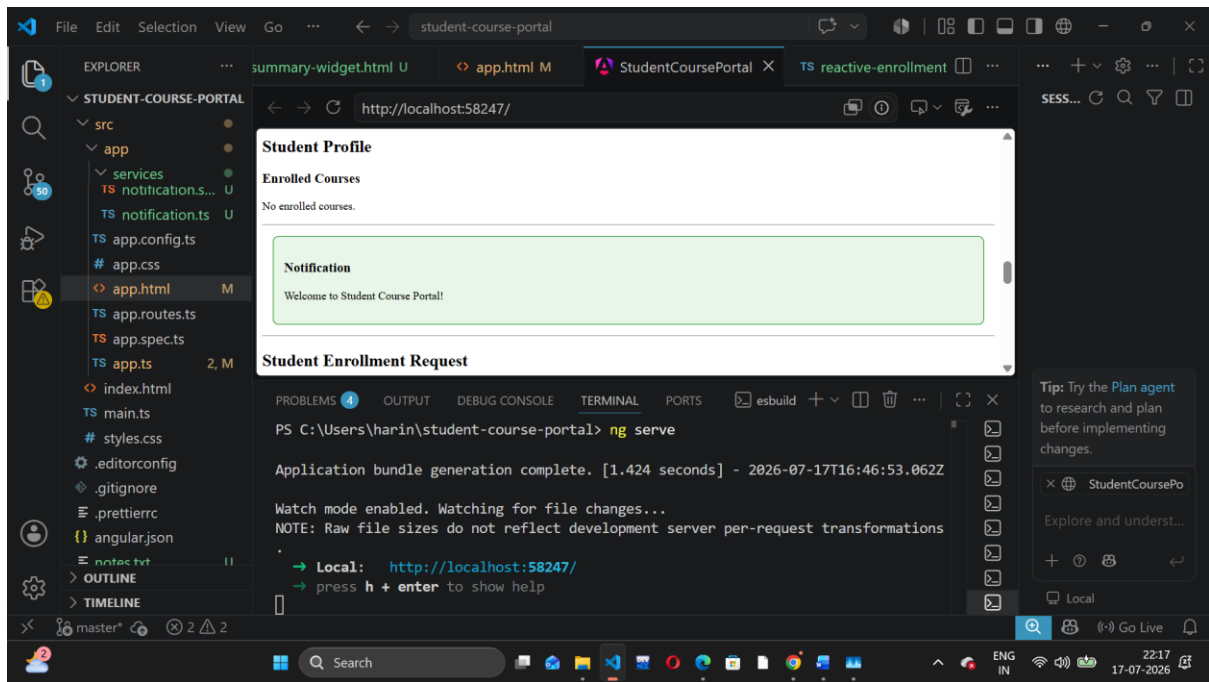
HANDS-ON 6 [Intermediate] Digital Nurture 5.0 | Angular (v20.0) | Hands-On Exercise Book ✓ Services & Dependency Injection

Task 1: Create and Use a Course Service Goal: Build a CourseService that provides course data to multiple components



Task 2: Enrollment Service and Hierarchical DI

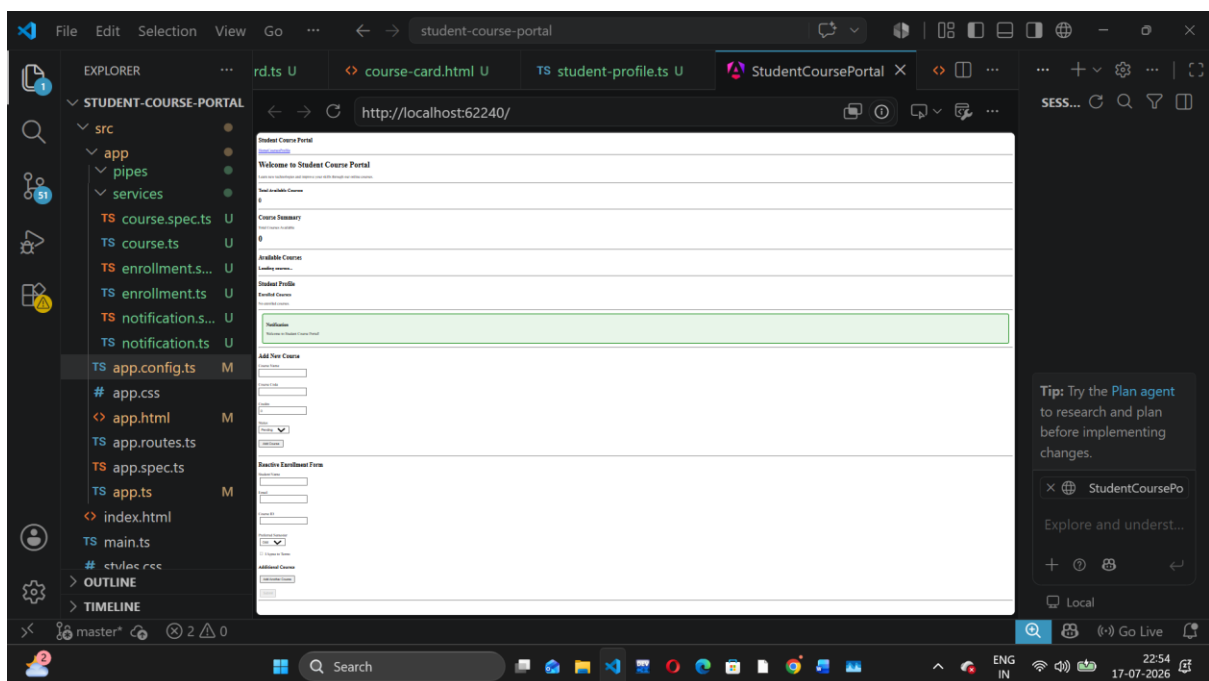
Goal: Create an EnrollmentService and demonstrate the DI hierarchy with component-level provider



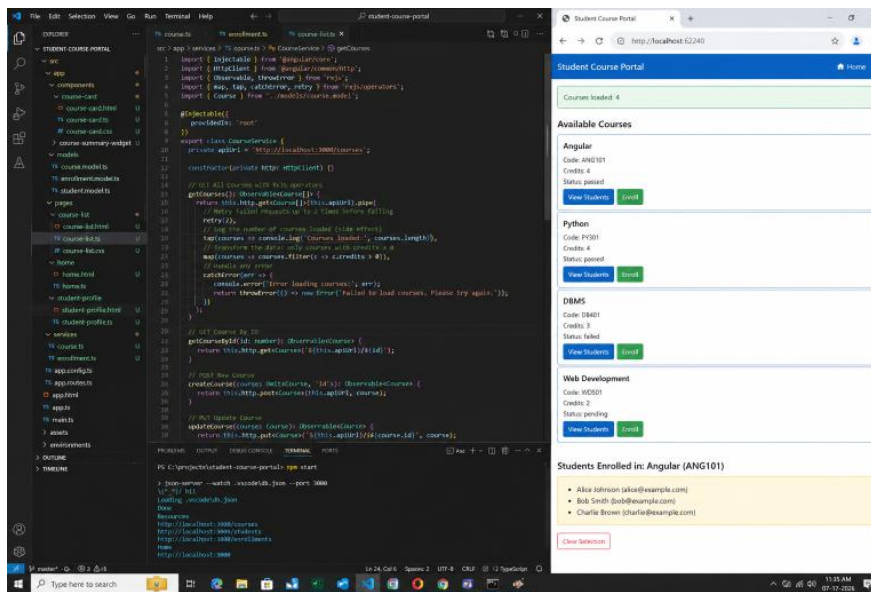
hy with component-level providers

AHNDS-ON 7[Advanced] Digital Nurture 5.0 | Angular (v20.0) | Hands-On Exercise Book ✓ HTTP Client — API Integration, Observables & Interceptors

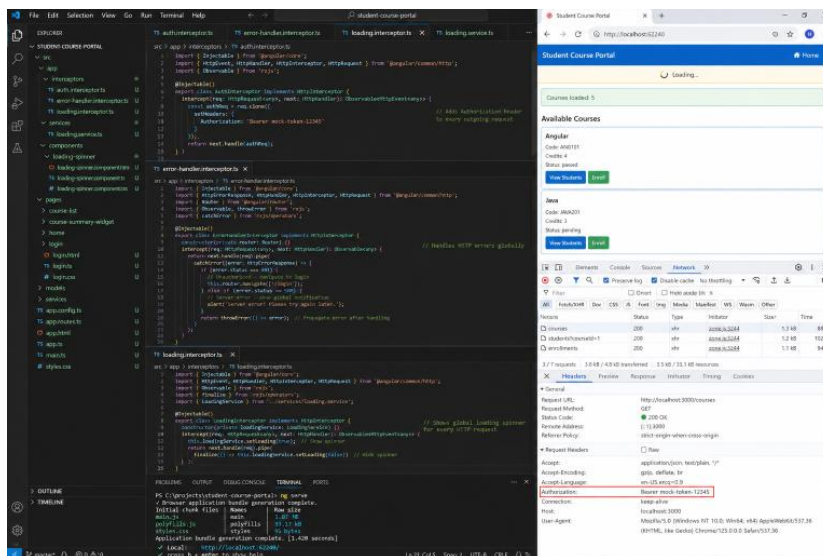
Task 1: Replace Service Data with HttpClient Calls Goal: Refactor CourseService and EnrollmentService to use real



Task 2: RxJS Operators and Error Handling Goal: Apply RxJS operators to transform and handle HTTP responses robustly.



Task 3: HTTP Interceptors Goal: Build interceptors to handle auth tokens and global error logging



HANDS-ON 9 [Advanced] Digital Nurture 5.0 | Angular (v20.0) | Hands-On Exercise Book ✓ State Management — NgRx Store, Actions, Reducers, Effects & Selectors

Task 1: Set Up NgRx Store and Define Course State Goal: Configure the NgRx store and create actions, reducer, and selectors for courses.


```
EXPLORE
STUDENT-COURSE-PORTAL
src
  app
    services
      TS course.service.spec.ts
      TS course-list.component.spec.ts
    components
    pages
    store
    app.config.ts
    app.routes.ts
    app.ts
    index.html
    main.ts
    styles.css
    test-setup.ts

TS course.service.spec.ts
1 import { TestBed } from '@angular/core/testing';
2 import { HttpClientTestingModule, HttpTestingController } from '@angular/common/http/testing';
3 import { CourseService } from '../services/course.service';
4 import { Course } from '../models/course.model';
5
6 describe('CourseService', () => {
7   let service: CourseService;
8   let httpMock: HttpTestingController;
9
10   beforeEach(() => {
11     TestBed.configureTestingModule({
12       imports: [HttpClientTestingModule],
13       providers: [CourseService]
14     });
15     service = TestBed.inject(CourseService);
16     httpMock = TestBed.inject(HttpTestingController);
17   });
18
19   it('should get courses', () => {
20     service.getCourses().subscribe((courses) => {
21       expect(courses).toHaveLength(3);
22     });
23   });
24
25   it('should get course by id', () => {
26     service.getById(1).subscribe((course) => {
27       expect(course).toEqual({
28         id: 1, name: 'Angular', credits: 3, gradeStatus: 'passed'
29       });
30     });
31   });
32
33   it('should handle error', () => {
34     service.getById(999).subscribe({
35       next: () => fail('should not reach here'),
36       error: err => {
37         expect(err.message).toContain('failed');
38       }
39     });
40   });
41 });
```

Karma v6.4.4 - connected

Chrome 124.0.0.0 (Windows 10) is idle

Finished in 1.94s

5 specs, 0 failures

Hide all

> CourseService

✓ should get courses

✓ should handle error

> CourseListComponent (NgRx)

✓ should render courses from store

✓ should show loading indicator when loading is true

Chrome 124.0.0.0 (Windows 10) Coverage

Generated: 17/07/2026, 11:25:44 am

AB Elements 50/52 96.15% Branches 15/16 93.75% Functions 10/10 100% Lines 50/52 96.15%

File Statements Branches Functions Lines

src/app/components/course-card 100% 100% 100% 100%

src/app/pages/course-list 94.44% 90% 100% 94.44%

src/app/services 95% 100% 100% 95%

src/app/store/course 97.3% 100% 100% 97.3%